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Investigation of Fluid Segregation and Phase Redistribution in Wellbore Pressure Buildup

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Summary

During wellbore pressure buildup tests, fluid segregation and phase redistribution commonly introduce oscillations and anomalies into pressure measurements that complicate Pressure Transient Analysis (PTA) interpretation. Fluid segregation and redistribution lead to three distinct pressure response types, with pronounced effects observed at high water cuts, where redistribution is most significant. Corrective techniques, such as rate normalization and type-curve matching, proved effective in mitigating the impact of redistribution on pressure analysis. However, these solutions fail to capture complex physics in the wellbore and cannot accurately represent phase redistribution effects at early times. This paper presents an integrated framework combining analytical solutions to overcome these challenges and improve reservoir characterization. We first review conventional type-curve techniques and identify their inability to capture early-time distortions. Building on the models of Fair (1981), Hegeman et al. (1993), and Li et al. (2014), we show that existing formulations cannot reproduce the varying timing of the pressure hump. To address these limitations, we derive a new analytical solution that incorporates a flexible time-delay term which allows the hump to be displaced in time and match field observations. We also introduce a cleanup strategy that removes phase redistribution effects directly from the measured pressure data rather than relying on pressure-adjustment fitting or matching in data affected by phase redistribution. A synthetic dimensionless example demonstrates that our approach accurately reproduces hump characteristics, recovers the ideal pressure response, and eliminates redistribution artifacts. The resulting framework equips engineers with practical tools for reliable data interpretation, improved reservoir parameter estimation, optimized well-testing procedures, and enhanced operational efficiency under multiphase flow conditions.

Introduction

Well testing is an important technology across oil and gas, groundwater and hydrogeology, carbon capture and storage, and geothermal applications. By analyzing flow rates and pressure behavior, well testing quantifies a well's productivity or injectivity efficiency and determines factors such as skin effects, deviations from Darcy flow, and wellbore storage. Well tests reveal key reservoir properties such as

connectivity and heterogeneity (for example, fractures or faults), prevailing flow regimes, the effective drainage area, and interactions between wells (Lee et al., 2003). In addition, pressure transient analysis solves the pressure-diffusivity equation. Therefore, analytical solutions emerge under simplified conditions (single-phase flow in a homogeneous medium with constant compressibility and well storage). However, to tackle more realistic reservoir geometries and fluid behaviors, engineers use type-curve matching techniques that employ dimensionless pressure and its derivative (A. Gringarten et al., 1979).

Pressure buildup tests remains a basis of reservoir and well performance evaluation across a wide range of subsurface applications including hydrocarbon extraction, carbon capture and storage (CCS), and geothermal energy (Bourdet, 2002; Lee et al., 2003). During the shut-in period, the behavior of bottom-hole pressure (BHP) encapsulates information of reservoir properties and wellbore dynamics. Under ideal single-phase, homogeneous conditions with constant compressibility and constant wellbore storage, classical pressure transient analysis (PTA) solves the radial diffusivity equation and employs dimensionless type-curve matching on pressure and its derivative to estimate skin factor, permeability, and other key parameters (A. C. Gringarten, 1987). However, as water cut increases and multiphase flow becomes prevalent within the wellbore, conventional interpretation face several challenges.

When fluid flow is stopped at surface, density contrasts between oil, water, and gas phases drive segregation within the wellbore. This process also known as phase redistribution gives rise to transient fluctuations in bottom-hole pressure measurements that may continue for hours or even days depending on fluid PVT properties, tubing configuration, and well trajectory (Pitzer et al., 1959; Stegemeier & Matthews, 1958). Such pressure fluctuations can appear as early-time "humps" or oscillations in semi-log buildup plots and lead to misleading straight-line intercepts in conventional Horner analyses (Horner, 1951). The practical consequence is that reservoir parameter estimates become biased unless the influence of phase segregation is properly identified and modeled.

The concept of wellbore storage was first introduced by Van Everdingen & Hurst (1949) and provided the foundation for understanding the concepts of afterflow, fluid compressibility effects, and phase redistribution during PTA. Field observations by Stegemeier & Matthews (1958) presented anomalous hump buildups in bottom-hole pressure records and highlighted the susceptibility of pressure-recorded curves to phase segregation and redistribution. From these observations, Pitzer et al. (1959) introduced specialized bottom-hole gauge techniques to minimize these distortions. Later, Fair (1981) introduced the first analytical model to capture phase redistribution as a time-dependent storage term in the diffusivity equation and demonstrated how even subtle fluid rearrangement can affect early-time data. The key assumptions of Fair (1981) include a constant gas/oil ratio in the wellbore, negligible temperature effects, ideal and incompressible behavior for both liquid and gas, and a linear increase in pressure with depth. Later, Thompson et al. (1986) extended the interpretation of such anomalous buildup tests by classifying buildup responses into three types of severity from substantial to negligible redistribution effects (Type I, II, and III). In addition, Rushing & Lee (1989) pioneered automatic history matching of buildup data impacted by wellbore phase segregation. Building on this, Olarewaju (1990) and used pressure-derivative analysis to distinguish segregation distortions characterized by a V-shaped depression from dual-porosity effects and the bell-shaped signature of wellbore storage. More recently, Mattar & Zaoral (1992) proposed the Primary Pressure Derivative (PPD), demonstrating that any reversal or non-monotonic trend in the PPD can clearly indicate wellbore segregation issues.

Subsequent developments have extended the treatment of changing wellbore storage and multiphase interactions. Hegeman et al. (1993) proposed an error-function formulation to capture both increasing and decreasing storage. Mechanistic models by Hasan & Kabir (1993) explored the influence of bubble rise velocity and wellhead pressure on transient response. Numerical-black-oil approaches by Xiao et al. (1996) introduced interphase mass transfer and variable bubble-point. Recent advances include oscillatory storage models that incorporate periodic correlations, new diagnostic plots for detecting low-level redistribution effects (Qasem et al., 2001, 2002), and physical storage models that account for under-saturated liquid and

gas dissolution dynamics (Gholamzadeh et al., 2020; Li et al., 2014). Despite these contributions, robust modeling and interpretation of phase redistribution in high water-cut wells remains a significant challenge.

In this work, we address the limitations of existing analytical models by deriving a revised solution that captures early-time phase redistribution behavior. The proposed analytical solution assumes radial flow of a slightly compressible fluid in an infinite, homogeneous, isotropic reservoir and is presented in dimensionless form. An additional term is introduced to control the timing of fluid-redistribution effects during pressure buildup tests. Moreover, our derivation demonstrates that these phase-redistribution effects can be removed analytically up front, eliminating the need for specialized type curves to reproduce the characteristic pressure humps. We then use this analytical expression with a numerical example under the influence of phase redistribution. This allows us to assess performance of the new analytical solution and approach to remove the phase redistribution effects. The results obtained confirm that our approach accurately reproduces the characteristic pressure hump seen in high water-cut wells and eliminates redistribution-induced distortions in buildup curves. By integrating analytical insight with a numerical solution, we provide a practical workflow that enables engineers to correct for segregation effects upfront and recover true reservoir behavior.

Methodology

Existing analytical models incorporate phase redistribution via a time-dependent storage term in the diffusivity equation. To obtain dimensionless pressure solutions that account for this effect during buildup tests, the radial diffusivity equation must be reformulated. Following Fair (1981) and assuming radial flow in an infinite, homogeneous, isotropic reservoir of slightly compressible fluid, the diffusivity equation in dimensionless form is expressed as:

$$\frac{\partial^2 p_D}{\partial r_D^2} + \frac{1}{r_D} \frac{\partial p_D}{\partial r_D} = \frac{\partial p_D}{\partial t_D} \quad (1)$$

Given the following boundary conditions:

$$p_D(r_D, 0) = 0 \quad (2)$$

$$\lim_{r_D \rightarrow \infty} p_D(r_D, t_D) = 0 \quad (3)$$

$$-\left(\frac{\partial p_D}{\partial r_D}\right)_{r_D=1} = 1 - C_D \left(\frac{dp_{wD}}{dt_D} - \frac{dp_{\phi D}}{dt_D} \right) \quad (4)$$

$$p_{wD} = \left[p_D - S \left(\frac{\partial p_D}{\partial r_D} \right) \right]_{r_D=1} \quad (5)$$

The solution of the diffusivity equation on the Laplace space is expressed as follows (where s denotes the Laplace transform variable):

$$\mathcal{A}(p_{wD}) = \frac{[s\mathcal{A}(p_D) + S] [1 + C_D s^2 \mathcal{A}(p_{\phi D})]}{s[1 + C_D s(s\mathcal{A}(p_D) + S)]} \quad (6)$$

Moreover, the required expression for $\mathcal{A}(p_D)$ provided by Van Everdingen & Hurst (1949) is a function of the Bessel functions of zero and first order.

$$\mathcal{A}(p_D) = \frac{K_0(\sqrt{s})}{s^{3/2} K_1(\sqrt{s})} \quad (7)$$

In the Laplace domain, phase redistribution appears through an extra $L(p_{\phi D})$ term in equation (6). Fair (1981) represents this term as a simple exponential increase in apparent storage. Hegeman et al. (1993) extend Fair's model with an error-function form that can describe increasing and decreasing storage regimes. Li et al. (2014) introduce a damped cosine term to capture the pressure oscillations associated with redistribution. Equations (8), (9), and (10) give the Laplace-space expressions for these three approaches. Fig. 1 compares the corresponding time-domain behaviors of each phase-redistribution model.

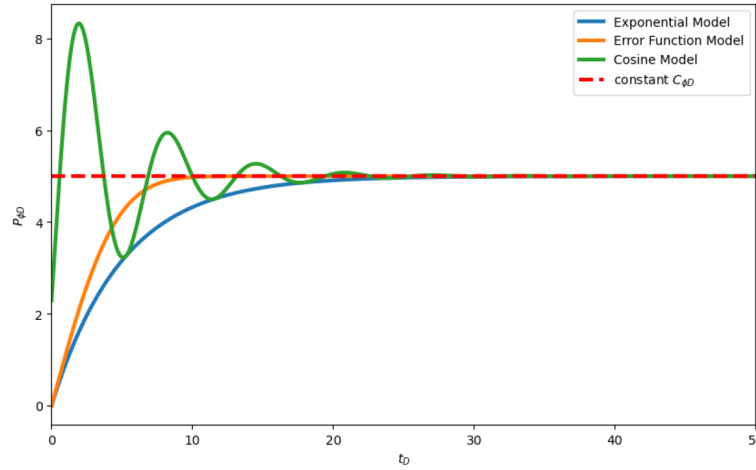


Figure 1—Comparative time-dependent analytical phase-redistribution expressions: exponential (Fair, 1981), error-function (Hegeman et al., 1993), and cosine (Li et al., 2014) formulations showing their characteristic pressure-increase responses and highlighting their main differences.

$$\mathcal{A}(p_{\phi D})_{\text{exp}} = \frac{C_{\phi D}}{s} - \frac{C_{\phi D}}{s + 1/\alpha_D} \quad (8)$$

$$\mathcal{A}(p_{\phi D})_{\text{eff}} = \frac{C_{\phi D}}{s} \exp(\alpha_D^2 s^2 / 4) \operatorname{erfc}(\alpha_D s / 2) \quad (9)$$

$$\mathcal{A}(p_{\phi D})_{\text{cos}} = C_{\phi D} \left(\frac{1}{s} - \frac{\left(\frac{s+1}{\alpha_D} \right) \cos \beta - \omega \sin \beta}{\left(\frac{s+1}{\alpha_D} \right)^2 + \omega^2} \right) \quad (10)$$

However, under varying wellbore storage conditions and dimensionless parameters, existing analytical models struggle to reproduce early-time pressure measurements. Consequently, type-curve matching heavily relies on middle- and late-time data.

New time-delay analytical expression

Phase redistribution effects occur mainly at early times, so an analytical solution is required to address this limitation directly. We, therefore, add an extra term to the error-function formulation (see equation (9)) in the Laplace domain since that form already captures both increasing and decreasing wellbore storage. The additional time delay term enables the analytical model to reproduce pressure increments caused by phase redistribution at different times. The new analytical formulation is presented in equation (11), and can be used in equation (6)

$$\mathcal{A}(p_{\phi D})_{\text{new}} = \exp(-(t_0 s)^\omega) \cdot \frac{C_{\phi D}}{s} \exp(s^2 / 4\alpha_{D2}) \operatorname{erfc}(s / 2\alpha_D) \quad (11)$$

Fig. 2 shows type curves generated for Dimensionless wellbore pressure p_{wD} by the new analytical formulation for phase redistribution at various time delays t_0 and dimensionless phase redistribution pressure

values $C_{\phi D}$. All cases use a wellbore storage coefficient of 100, skin factor of 1, dimensionless time parameter α_D of 0.1, and dimensionless frequency ω of 0.4. The new expression allows independent control over when phase redistribution occurs and the magnitude of the resulting pressure hump. This new time delay analytical solution could now be used to perform conventional Type-Curve analysis and obtain the underlying reservoir characteristics of interest, such as permeability and skin.

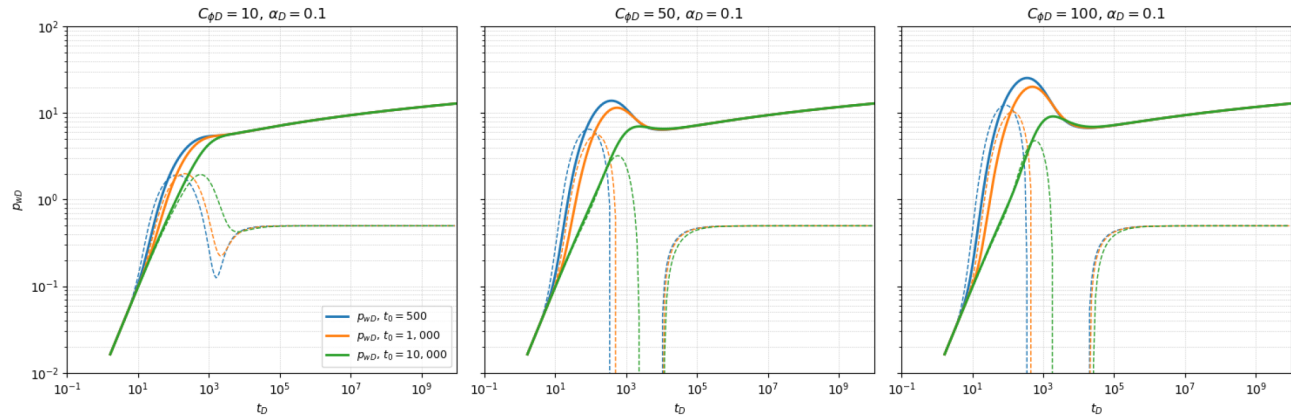


Figure 2—Dimensionless wellbore pressure p_{wD} generated from the new time-delay analytical expression for various time delays and dimensionless redistribution pressure coefficients $C_{\phi D}$. The generated curves demonstrate that the added term effectively models phase-redistribution effects across varying time scales.

Eliminating Phase Redistribution Effects Up Front

From the dimensionless wellbore pressure solution given in [equation \(6\)](#), the term p_{ϕ} explicitly captures the effects of phase redistribution. Consequently, the Laplace-domain expression for the dimensionless wellbore pressure when no effects of phase redistribution are present, denoted as $L(p_{wD})_{ideal}$, can be written as:

$$\mathcal{A}(p_{wD})_{ideal} = \frac{s\mathcal{A}(p_D) + S}{s[1 + C_D s(s\mathcal{A}(p_D) + s)]} \quad (12)$$

Next, we rearrange the solution so that the Laplace-domain wellbore pressure, $L(p_{wD})$, is expressed in terms of the clean (ideal) pressure and the redistribution contribution:

$$\mathcal{A}(p_{wD}) = \mathcal{A}(p_{wD})_{ideal} [1 + C_D s^2 \mathcal{A}(p_{\phi D})] \quad (13)$$

Finally, solving [equation \(13\)](#) for $L(p_{wD})_{ideal}$, yields the Laplace transform of the pressure signal with all phase redistribution effects removed, where $L(p_{wD})$ in this context represents the phase redistribution-affected pressure.

$$\mathcal{A}(p_{wD})_{ideal} = \mathcal{A}(p_{wD}) \frac{1}{[1 + C_D s^2 \mathcal{A}(p_{\phi D})]} \quad (14)$$

To solve [equation \(14\)](#), we need the recorded wellbore pressure in the dimensionless form that contains phase redistribution effects. We compute its Laplace transform by numerically integrating each pressure sample multiplied by an exponential decay factor over the time array to produce a Laplace-domain representation that still reflects phase redistribution. We then solve [equation \(14\)](#) in Laplace space using the Stehfest inversion algorithm to recover the wellbore pressure in the time domain without any phase redistribution distortion.

Results

To test the capabilities of the proposed formulation, we constructed a fully synthetic, dimensionless test case to evaluate the new analytical formulation (equation (11)) and proposed framework. The chosen parameter values are: $\alpha_D=0.1$, $C_{\phi D}=50$, $t_0=10\,000$, $\omega=0.1$, $C_D=100$, and $S=1$. These values define the dimensionless wellbore pressure p_{wD} affected by phase redistribution. Using Equation (12) and the same parameter set, we then construct the corresponding clean (ideal) pressure signal. In this way, we obtain both the distorted and the ideal pressure curves (see Figure Fig. 3, left panel). Finally, by solving Equation (14) as described in the previous section, we remove the phase-redistribution contribution and recover the corrected pressure response (Fig. 3, right panel).

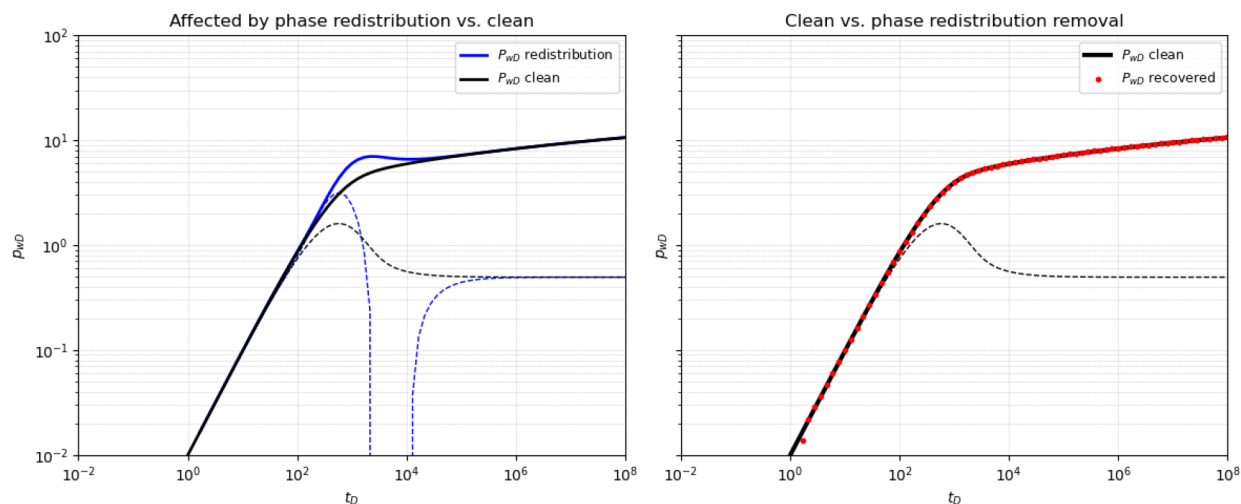


Figure 3—Example of the phase redistribution removal proposed. Left panel: Dimensionless wellbore pressure calculated using the new analytical expression overlaid with the corresponding clean signal (pressures and derivatives). Right panel: Reconstructed signal from the pressure measurements affected by phase redistribution.

Discussion

This study introduces a new analytical solution for modeling how phase redistribution distorts wellbore pressure during buildup tests. Building on the work of Fair (1981), Hegeman et al. (1993), and Li et al. (2014), we extend Hegeman's formulation since it captures both increasing and decreasing storage. Our revised model resolves inconsistencies in existing early-time solutions by adding a flexible time delay, which shifts the pressure hump to match field observations. Conventional type curve analysis and history matching in pressure transient analysis remain valid when using this model. We also propose an alternative strategy for handling phase redistribution: instead of fitting the redistribution signal, we remove its contribution directly from measured pressure data. This cleanup approach can be applied to any published analytical forms, including the new solution, and yields a pressure profile that reflects only the true reservoir response. The synthetic example shows that this method successfully eliminates the distorted hump and recovers the ideal pressure curve. Future work should develop more robust techniques for removing these artifacts and evaluate the impact on reservoir parameter estimation in complex field settings. Although we have introduced a novel analytical solution and an upfront method for eliminating phase-redistribution effects, the inverse Laplace transform of equation (14) remains nontrivial in practice. In particular, the Stehfest algorithm (popular for its simplicity) frequently faces stability and convergence issues when dealing with the presented analytical solution to remove segregation effects. This limitation highlights the need to investigate more sophisticated inversion schemes. One practical workaround is to pre-process the pressure data in the time domain into the Laplace space. Finally, practitioners can leverage the new dimensionless formulation

to generate an updated suite of type curves. Therefore, users can achieve more reliable type curve-matching, reduce interpretational ambiguity, and improve estimates of key properties such as permeability and skin.

Conclusions

This work has presented a new analytical solution that accurately models wellbore pressure distortions caused by phase redistribution during buildup tests. By extending Hegeman's formulation to include a flexible time-delay term, the revised model reproduces early-time pressure humps at the correct interval rather than forcing all distortions into a single fixed shape. This advancement preserves the applicability of type-curve matching and history-matching workflows in pressure transient analysis while improving the fidelity of early-time predictions. Additionally, we introduced a direct cleanup strategy that removes the phase redistribution contribution in measured pressure data. This approach can be applied to any existing analytical form and yields a pressure profile that reflects only reservoir behavior. Synthetic examples demonstrate that the cleanup method successfully recovers the ideal pressure response, eliminating artifacts without requiring case-specific tuning of the hump shape. Future research should focus on validating these findings against field-acquired buildup tests in high water-cut or complex well geometries. Further improvements may come from combining the time-delay formulation with more complex physics of the fluid flow within the wellbore. Ultimately, more robust artifact-removal techniques will strengthen reservoir parameter estimates and enhance the reliability of pressure transient analysis under multiphase flow conditions. Moreover, this work suggests a promising avenue for enhancing the reliability of pressure buildup tests in high water-cut wells by systematically removing phase-segregation and redistribution artifacts from the recorded pressure data. In this manner, we envision purely data-driven strategies such as machine-learning frameworks that can recover the true reservoir response directly from field data without relying on a closed-form analytical solution. Such approaches could leverage deep denoisers to learn and subtract the undesired pressure distortions and ultimately improve the accuracy of inferred reservoir properties.

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